Computer Programming 1

Exercises on Basic Concepts

(16-09-2026)

Exercise 1: Social Distance (task)

Context: You are in a room, and you want to be sure that you have enough space around you to distance yourself from other persons.

Task: write a simple C++ program with that:

1. assigns to a variable *r* the measure of a radius expressed by a real number,
2. computes the area (A) and the circumference (C) of a circle by using:

$$C = 2\pi r \qquad A = \pi r^2$$

3. prints in the screen: the value of the radius, the computed area and the computed circumference

Constraint: use float as datatype of the radius variable

Exercise 2: American thermometer (task)

Context

You have been instructed to take the temperature of all participants in the programming course of the University. Unfortunately, the supplied thermometer is set only in Fahrenheit!

Task: write a simple C++ program with that:

1. defines a variable with the value of the temperature in Fahrenheit,
2. computes the corresponding value of the temperature in Celsius degrees by using the following formula:

$$T_C = \frac{T_F - 32}{1.8}$$

Exercise 3: Convert (task)

Task

Given a set of alphanumeric characters (i.e., letters and numbers) stored in different variables (one variable for one character), the program needs to convert all uppercase characters (letters) by returning their lowercase counterpart. Print in the monitor the converted characters.

Constraint: do not use any libraries or function to automatically convert uppercase to lowercase characters

Suggestion: use the ASCII table (see next slide)

ASCII table

Standard ASCII Chart / ASCII Table - Hex to Decimal Code Conversion

Dec	Hex	Name	Char	Ctrl-char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
0	0	Null	NUL	CTRL-@	32	20	Space	64	40	@	96	60	`
1	1	Start of heading	SOH	CTRL-A	33	21	!	65	41	A	97	61	a
2	2	Start of text	STX	CTRL-B	34	22	"	66	42	B	98	62	b
3	3	End of text	ETX	CTRL-C	35	23	#	67	43	C	99	63	c
4	4	End of xmit	EOT	CTRL-D	36	24	$	68	44	D	100	64	d
5	5	Enquiry	ENQ	CTRL-E	37	25	%	69	45	E	101	65	e
6	6	Acknowledge	ACK	CTRL-F	38	26	&	70	46	F	102	66	f
7	7	Bell	BEL	CTRL-G	39	27	'	71	47	G	103	67	g
8	8	Backspace	BS	CTRL-H	40	28	(	72	48	H	104	68	h
9	9	Horizontal tab	HT	CTRL-I	41	29	)	73	49	I	105	69	i
10	0A	Line feed	LF	CTRL-J	42	2A	*	74	4A	J	106	6A	j
11	0B	Vertical tab	VT	CTRL-K	43	2B	+	75	4B	K	107	6B	k
12	0C	Form feed	FF	CTRL-L	44	2C	,	76	4C	L	108	6C	l
13	0D	Carriage feed	CR	CTRL-M	45	2D	-	77	4D	M	109	6D	m
14	0E	Shift out	SO	CTRL-N	46	2E	.	78	4E	N	110	6E	n
15	0F	Shift in	SI	CTRL-O	47	2F	/	79	4F	O	111	6F	o
16	10	Data line escape	DLE	CTRL-P	48	30	0	80	50	P	112	70	p
17	11	Device control 1	DC1	CTRL-Q	49	31	1	81	51	Q	113	71	q
18	12	Device control 2	DC2	CTRL-R	50	32	2	82	52	R	114	72	r
19	13	Device control 3	DC3	CTRL-S	51	33	3	83	53	S	115	73	s
20	14	Device control 4	DC4	CTRL-T	52	34	4	84	54	T	116	74	t
21	15	Neg acknowledge	NAK	CTRL-U	53	35	5	85	55	U	117	75	u
22	16	Synchronous idle	SYN	CTRL-V	54	36	6	86	56	V	118	76	v
23	17	End of xmit block	ETB	CTRL-W	55	37	7	87	57	W	119	77	w
24	18	Cancel	CAN	CTRL-X	56	38	8	88	58	X	120	78	x
25	19	End of medium	EM	CTRL-Y	57	39	9	89	59	Y	121	79	y
26	1A	Substitute	SUB	CTRL-Z	58	3A	:	90	5A	Z	122	7A	z
27	1B	Escape	ESC	CTRL-[	59	3B	;	91	5B	[	123	7B	{
28	1C	File separator	FS	CTRL-\	60	3C	<	92	5C	\	124	7C	|
29	1D	Group separator	GS	CTRL-]	61	3D	=	93	5D	]	125	7D	}
30	1E	Record separator	RS	CTRL-^	62	3E	>	94	5E	^	126	7E	~
31	1F	Unit separator	US	CTRL-_	63	3F	?	95	5F	_	127	7F	DEL

Exercise 4: seconds to midnight

- Write a program that given three numbers (hours, minutes and seconds), stored in three distinct variables, and then calculates the seconds to midnight. Print the value of the three variables and the computed seconds that remain to midnight.

- E.g., hours:23 minutes:59 seconds:59 => 1
- E.g., hours:22 minutes:15 seconds:2 => 6298

Exercise 5: seconds to midnight (v2)

- Extend the previous program (i.e., “seconds to midnight”) by writing a new program that implements also the following request:
  - given the seconds to midnight, stored in a variable, it returns the hourly equivalent in hours, minutes and seconds to the screen.

- E.g., seconds to m. = 1 => hours:0 minutes:0 seconds:1
- Eg., seconds to m. = 6298 => hours:1 minutes:44 seconds:58

Additional Exercises

- Variant #1. Given the radius of a circle as input, the program needs to calculate the volume of the corresponding sphere

- Variant #2. Define two variables with two numbers (dividend and divisor), the program needs to calculate the quotient and the remainder of the division operation